Kong

WEBINAR · KONG AI GATEWAY

Migrating to Kong AI Gateway 2.0

Models, providers and agents you declare once

Why move, what changes, how to migrate safely, and what to watch for. With a live migration from Kong Gateway 3.16 to Kong AI Gateway 2.3.

SPEAKER MARAT SALAKHUTDINOV **DATE** NOVEMBER 12, 2026 **FORMAT** LIVE DEMO

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AGENDA

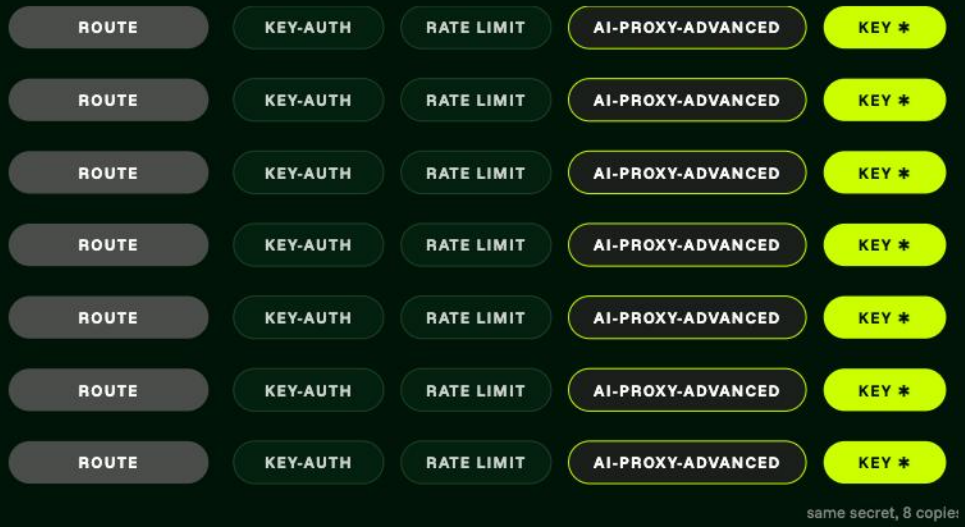
Forty-five minutes, one migration

1	The AI platform we built on 3.x	3 min
2	Why migrate	5 min
3	From plugins to entities	5 min
4	The migration approach	5 min
5	Live demo	20 min
6	Lessons from the field	4 min
7	Checklist, next steps, Q&A	3 min +

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WHERE MOST OF US ARE

The AI platform we all built on 3.x



- One service and route per model or API
- Each route: its own `ai-proxy-advanced`, auth and rate limiting
- The provider credential repeated on every route
- AI intent spread across many primitives

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THE DEMO · KONG GATEWAY 3.16

Our starting point

ROUTE	PROVIDER / MODEL	API
`/openai/v1/chat/completions`	OpenAI gpt-5-mini	OpenAI chat
`/openai/v1/responses`	OpenAI gpt-5-mini	Responses
`/openai/v1/embeddings`	OpenAI text-embedding-3-small	Embeddings
`/bedrock/v1/chat/completions`	Bedrock · Claude Haiku 4.5	OpenAI format
`/anthropic/v1/messages`	Bedrock · Claude Haiku 4.5	Anthropic Messages
`/vertex/v1/chat/completions`	Vertex AI · Gemini 2.5 Flash	OpenAI format
`/vertex/v1/embeddings`	Vertex gemini-embedding-001	Embeddings
`/gemini/…:generateContent`	Vertex AI · Gemini 2.5 Flash	Native Gemini

APPLICATIONS · API KEYS

CONSUMER	TIER	TOKENS/MIN
app-reporting	gold	50,000
app-chatbot	silver	5,000
app-batch	bronze	300
app-etl	bronze	300

Keys in `apikey`, or the header each SDK sends: `x-api-key`, `x-goog-api-key`.

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TODAY · KONG GATEWAY 3.16

The current setup: one plugin stack per route

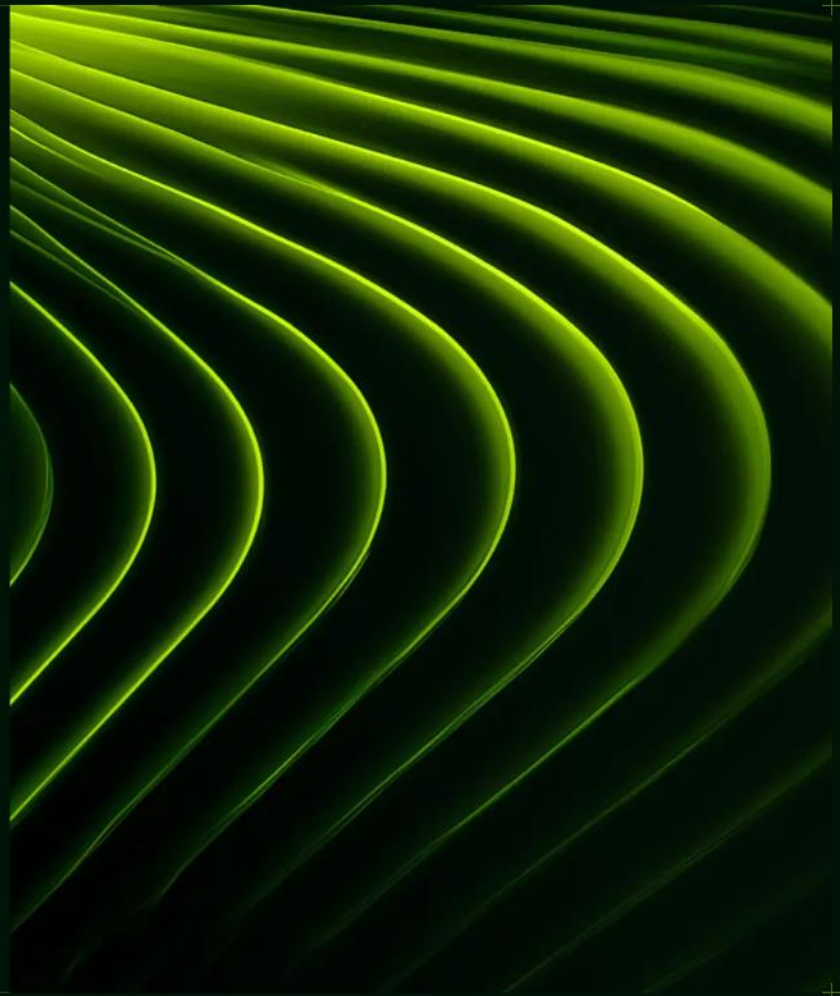


1 key-auth · 3 consumer groups, 3 ai-rate-limiting-advanced · 3 services · 8 routes, 8 ai-proxy-advanced · 3 provider keys copied 8 times

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SECTION 2

Why move to Kong AI Gateway 2.x



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WHY NOW

One codebase, two release trains

The AI Gateway ships on its own cadence, independent of the API Gateway.

WHAT

Two independent release trains

A dedicated AI Gateway runtime: `kong/kong-ai-gateway`.

WHERE

Konnect first

Where most customers already run their gateways, with a dedicated AI Gateway control plane.

WHY

Move at AI speed

Ship faster, take on bigger runtime changes, and iterate on AI-native UX, with zero risk to API management customers.

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WHAT SHIPS

What Kong AI Gateway 2.x brings

1

A dedicated AI Gateway runtime

A purpose-built data plane, released on its own train.

2

A streamlined Konnect UI

A dedicated entity model, with first-class support for LLMs, MCP servers and agents.

3

Declarative with kongctl

Including the migration tooling from Kong Gateway 3.x: `kongctl convert ai-gateway`.

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THE BIG IDEA

From plugins to entities

You no longer assemble services, routes and plugins by hand. You declare the AI resource you want, and the control plane provisions the proxying for you.

AI GATEWAY V1 · PLUGIN MODEL

- Build a service and route, then attach a plugin: `ai-proxy-advanced`, `ai-mcp-proxy`, `ai-a2a-proxy`
- One logical model = a service + route + plugin + supporting plugins
- AI intent is spread across many primitives



AI GATEWAY V2 · NATIVE ENTITY MODEL

- Declare the AI resource; the control plane wires the proxying
- First-class entities on a dedicated AI Gateway control plane
- One entity per logical model, MCP server or agent
- AI intent lives in one place

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CORE MODEL

The new entity model

NEW FIRST-CLASS ENTITIES

- MODELS
- MODEL PROVIDERS
- MCP SERVERS
- AGENTS (A2A + HTTP)
- POLICIES

CARRIED OVER FROM KONG 3.X

- CONSUMERS
- CONSUMER GROUPS
- VAULTS
- CONFIG STORES
- AUTH STRATEGIES

- Models are implemented by models + providers
- Plugins become policies
- Everything is managed under one AI Gateway control plane: Konnect UI, API and kongctl

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MODELS + PROVIDERS

One plugin becomes two reusable entities

V1 · ONE AI-PROXY-ADVANCED PLUGIN

- Provider, credentials, route type, model options and load balancer, all in one place
- Credentials inlined per target
- Reuse means copy and paste across plugins

→

PROVIDER

Upstream connection + credentials. Declared once, referenced by any number of models.

MODEL

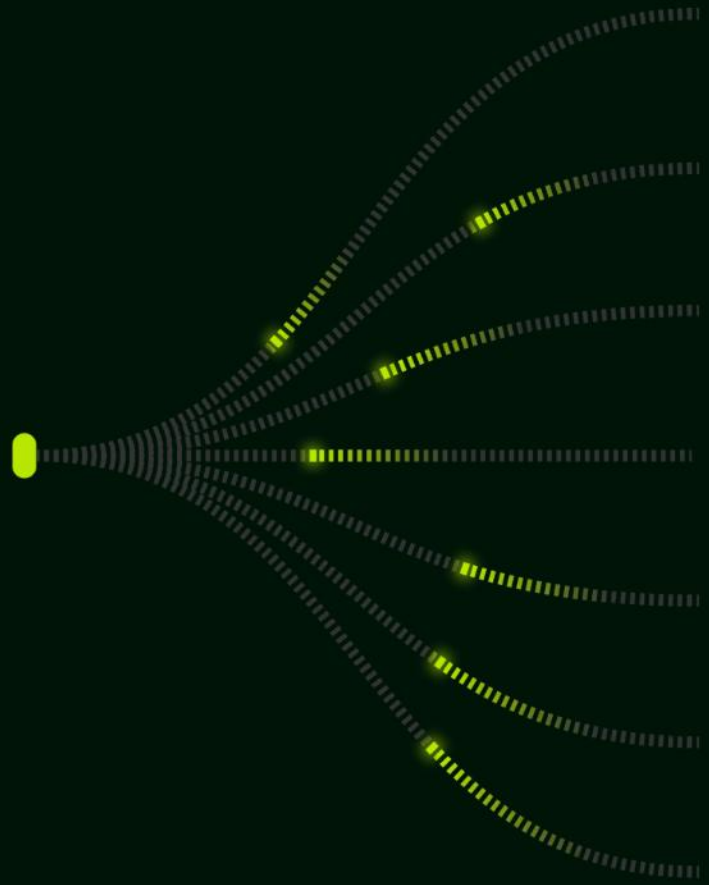
Routing, capabilities, format, balancing, and targets that each reference a provider.

Call it out: `route_type` splits into a capabilities list + an API formats entry: `llm/v1/chat` becomes `capabilities: [generate]` + `formats: [openai]`.

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SECTION 3

How 3.x maps to 2.x



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ENTITY MAPPING

3.x concepts, 2.x entities

KONG GATEWAY 3.X	KONG AI GATEWAY 2.X
route + `ai-proxy-advanced` target	**Model** · route paths kept; `route.model.values` selects on the body `model`
credential inside each target	**Model provider** · one per credential, referenced by models
`route_type: llm/v1/chat`	`capabilities: [generate]` + `formats: [openai]`
global `key-auth` plugin	**Auth strategy** (key-auth) attached to models
consumer + key-auth credential	**Consumer** + credential, same key
consumer group + scoped `ai-rate-limiting-advanced`	**Consumer group** + **policy**

Source: v1 to v2 table, developer.konghq.com/ai-gateway/ai-gateway-v2-concepts

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TWO CLIENT-VISIBLE RULES

What 2.x expects from a request

1

The route path is a base path

The gateway appends the capability path: `/chat/completions`, `/embeddings`, `/responses`, `/v1/messages`.

2

The request picks the model

The body `model` (or the model in the Gemini URL) must match the model's `route.model.values`.

The converter handles both: paths become base paths, 3.x model names become selectors. **But a request without `model` gets 503 on 2.x**, where 3.x used the route's pinned model.

ONE ROUTE, BEFORE AND AFTER

Same intent, platform shape

3.16 · DECK

```
- name: openai-chat
  paths: [/openai/v1/chat/completions]
  plugins:
    - name: ai-proxy-advanced
      config:
        targets:
          - route_type: llm/v1/chat
            auth:
              header_name: Authorization
              header_value: '${{ env "DECK_OPENAI_AUTH" }}'
            model:
              provider: openai
              name: gpt-5-mini
```

2.3 · KONGCTL

```
ai_gateway_model_providers:
  - name: openai-1
    type: openai
    config:
      auth:
        headers:
          - name: Authorization
            value: !secret {source: !env DECK_OPENAI_AUTH}
ai_gateway_models:
  - name: openai-chat
    formats: [{type: openai}]
    targets: [{name: gpt-5-mini, provider: openai-1}]
    access: {auth_strategies: [api-key]}
    config:
      route:
        paths: [/openai/v1]
        model: {body_param: model, values: [gpt-5-mini]}
```


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AFTER · KONG AI GATEWAY 2.3

After the migration: declared once, referenced everywhere



1 auth strategy · 3 consumer groups, 3 policies · 8 models, no plugins · 3 model providers, 1 key each

ROOM TO IMPROVE Models that share an upstream model can become one model with several capabilities: openai-chat + openai-responses → generate + agentic

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SECTION 4

The migration, as a repeatable pipeline



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THE APPROACH

Six steps from 3.x to 2.x

1

Prepare and export

Tiers in the `policies` format (3.14+), then `deck gateway dump`; count routes, plugins, consumers.

2

Convert

`kongctl convert ai-gateway` with a small `config/` folder.

3

Review and fix

Read `migration-report.md`; apply the fixes on a copy.

4

Plan and apply

`kongctl plan --mode apply`, then `kongctl apply`. Never `sync` shared namespaces.

5

Run in parallel

Same requests and keys against both gateways; compare.

6

Cut over

Switch DNS or the load balancer; keep 3.x warm as the rollback.

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WHAT THE CONVERTER NEEDS FROM YOU

A small `config/` folder

- `gateway.yaml`: the target AI Gateway. For an existing gateway, run `kongctl adopt ai-gateway` once.
- `auth_strategies.yaml`: how global auth plugins become strategies. Only one strategy may use `models: ['*']`.
- Vault entities only if your vault is a Konnect entity.

TOOLING USED TODAY

kongctl	`1.28.0`
converter extension	`v0.6.0`
decK	`1.70.0`
Kong Gateway	`3.16.0.2`
Kong AI Gateway	`2.3.1`

```
kongctl install extension \
  Kong/kongctl-ext-aigw-converter --yes
```

SECTION 5 · LIVE

Live migration: 3.16 to 2.3

1 · THE BEFORE
2 · APPS AND KEYS
3 · EVERY API
4 · CONVERT AND REVIEW
5 · THE AFTER
6 · BOTH GATEWAYS
7 · RATE LIMITS, THE RIGHT WAY

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SCENE 6 · SAME REQUESTS, BOTH GATEWAYS

15 of 15 checks, on both

CHECK	3.16	2.3
No key / unknown key → 401	✓	✓
OpenAI chat, responses, embeddings	✓	✓
Bedrock Claude, OpenAI format + Anthropic Messages	✓	✓
Vertex Gemini, OpenAI format + embeddings + native	✓	✓
SDK headers `x-api-key`, `x-goog-api-key`	✓	✓
Bronze 429, gold never limited, one budget per consumer	✓	✓

SAME KEYS, SAME URLS, SAME HEADERS

```
# 429 on both gateways
x-ai-ratelimit-limit-minute-policy-1: 300
x-ai-ratelimit-remaining-minute-policy-1: 0
x-ai-ratelimit-retry-after: 46
```

The rate-limit headers are identical on both gateways, so clients that pace themselves need no change.

LESSONS FROM THE FIELD

The converter does most of the work

Routes become models, credentials become providers, consumers keep their keys, and tiers carry over as consumer groups with policies. A report lists what clients must send and anything it couldn't map.

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LESSONS FROM THE FIELD

Review, test, and ask for help when needed

Most items convert as they are. Where something doesn't, a few manual steps may be needed: read `migration-report.md`, send the same requests to both gateways, and reach out to Kong support or your CSM for anything the converter can't map.

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SECTION 7

Migration checklist

- [ ] Move 3.x tiers to the `policies` format, with `partition_by`; test on 3.x
- [ ] Dump every control plane or workspace with AI routes, consumers and groups
- [ ] Write `config/`: gateway, auth strategies
- [ ] Convert, read `migration-report.md`, fix on a copy
- [ ] Plan, review, apply to a non-production AI Gateway
- [ ] Same vault access and TLS settings on the 2.x data plane
- [ ] Same test suite on both, including per-consumer budgets and requests without `model`
- [ ] Tell clients to always send `model`; agree a cutover window
- [ ] Switch traffic; keep 3.x warm until error rates and usage match

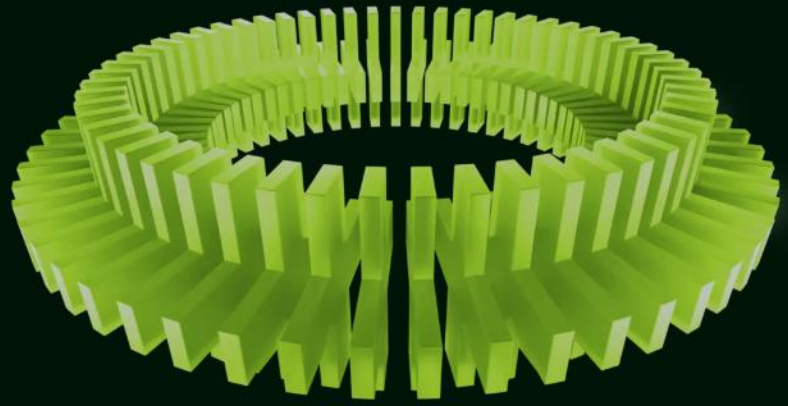
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TAKE-HOME KIT

Run the same migration on your gateways

- `migration-runbook.html`: fill in your settings, every command is ready to copy
- Starts from a 3.x control plane and an AI Gateway that already run
- Config examples for every stage, from the 3.x file to what gets applied
- No identity provider and no data plane settings needed

Download the kit and the recording: `konghq-cx.github.io/cs-webinars`



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REFERENCES

Bookmark these

TOPIC	LINK
Migration guide	developer.konghq.com/ai-gateway/v2-migration-guide
Entity mapping (v1 → v2)	developer.konghq.com/ai-gateway/ai-gateway-v2-concepts
Kong AI Gateway docs	developer.konghq.com/ai-gateway
Changelog	developer.konghq.com/ai-gateway/changelog
Converter	github.com/Kong/kongctl-ext-aigw-converter
kongctl	developer.konghq.com/kongctl · github.com/Kong/kongctl
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